

Machine Learning

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Lecture 07: Performance Measures

Sources

- Jurafsky and Martin, SLP3,
<https://web.stanford.edu/~jurafsky/slp3/>

More than two classes

Lots of classification tasks in language processing have more than two classes:

- Sentiment analysis (positive, negative, neutral),
- Part-of-speech tagging (|POS tags|)
- Word sense disambiguation (|word senses|)
- Emotion detection (|emotions|)

More Than Two Classes: Sets of binary classifiers

Any-of or multi-label classification

- An instance can belong to 0, 1, or more than one classes.

For each class $c \in C$

- Build a binary classifier h_c to distinguish c from all other classes $c' \in C$ (c vs not c)

Given test instance d ,

- Evaluate it for membership in each class using each h_c
- d belongs to any class for which h_c returns true

More Than Two Classes: Sets of binary classifiers

One-of or multinomial classification

- Classes are mutually exclusive: each instance in exactly one class

For each class $c \in C$

- Build a classifier h_c to distinguish c from all other classes $c' \in C$.

Given test instance d ,

- Evaluate it for membership in each class using each h_c
- d belongs to the one class with maximum score

Evaluation

- 3-way one-of email categorization decision (urgent, normal, spam)

		<i>gold labels</i>			
		urgent	normal	spam	
<i>system output</i>	urgent	8	10	1	$\text{precision}_u = \frac{8}{8+10+1}$
	normal	5	60	50	$\text{precision}_n = \frac{60}{5+60+50}$
	spam	3	30	200	$\text{precision}_s = \frac{200}{3+30+200}$
		$\text{recall}_u = \frac{8}{8+5+3}$	$\text{recall}_n = \frac{60}{10+60+30}$	$\text{recall}_s = \frac{200}{1+50+200}$	

Figure 6.5 Confusion matrix for a three-class categorization task, showing for each pair of classes (c_1, c_2) , how many documents from c_1 were (in)correctly assigned to c_2

Per class evaluation measures

- **Recall:**
 - Fraction of instances in class i classified correctly:
- **Precision:**
 - Fraction of instances assigned class i that are actually about class i :
- **Accuracy: (1 - error rate)**
 - Fraction of instances classified correctly:

Micro- vs. Macro-Averaging

- If we have more than one class, how do we combine multiple performance measures into one quantity?
- **Macro-averaging:** Compute performance for each class, then average.
- **Micro-averaging:** Collect decisions for all classes, compute contingency table, evaluate

Evaluation

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Figure 6.5 Confusion matrix for a three-class categorization task, showing for each pair of classes (c_1, c_2), how many documents from c_1 were (in)correctly assigned to c_2

Class 1: Urgent

true
urgent

true
not

system
urgent

system
not

8

11

8

340

precision = $\frac{8}{8+11}$ = .42

macroaverage
precision = $\frac{.42+.52+.86}{3}$ = .60

Class 2: Normal

true
normal

true
not

system
normal

system
not

60

55

40

212

precision = $\frac{60}{60+55}$ = .52

Class 3: Spam

true
spam

true
not

system
spam

system
not

200

33

51

83

precision = $\frac{200}{200+33}$ = .86

Pooled

true
yes

true
no

system
yes

system
no

268

99

99

635

microaverage
precision = $\frac{268}{268+99}$ = .73

Evaluation

- A micro-average is dominated by the more frequent class (in this case spam)
 - as the counts are pooled
- The macro-average better reflects the statistics of the smaller classes,
 - is more appropriate when performance on all the classes is equally important.

Thank You